

Pioneering vascular science to treat intractable diseases

Corporate Presentation
June 2025

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Focused on Innovative Therapeutics Through Vascular Science

Company Overview



Name	CURACLE Co.,Ltd.
Established	May 2016, Seoul, South Korea
Listed	July 2021 (Technology Evaluation: AA, Grade A)
Number of Employees	34 (21 in R&D, 62% in total workforce dedicated to R&D)
Research Centers	Central Research / Bio Research / New Drug Research Center

R&D Focus



Area of Development	Intractable diseases caused by vascular endothelial dysfunction
Core Technology	Drug development platform <i>SOLVADYS®</i> & <i>EAGLES®</i>
Pipeline Portfolio*	16 First-in-Class pipelines (8 Small molecules + 8 Biologics)
Pipeline Development Phases**	Discovery > Preclinical > Phase 1 > Phase 2 & later 6 3 2 5

*The number of pipelines classified by new substances or indications is 16. When classified solely by new substances, the number is 11.
** As of October 2024, this includes the number of projects currently in each stage, or those that have completed the previous stage and are about to enter the next stage (including joint research projects through open innovation, excluding projects delayed or given lower priority).

1. Company Introduction

20+ Years of Dedicated Vascular Research

Founder



Young-Guen Kwon

- Expert in Vascular Diseases
(SCI Papers 225, Registered Patents 15, L/O 3)

Education and Career

- Professor at Yonsei University
- Dean of Yonsei Univ. College of Life Science & Biotechnology
- President of the Korean Society for Vascular Biology and Medicine
- Advisory Member, Presidential Advisory Council on Science and Technology
- Member, The Korean Academy of Science and Technology
- Full-time researcher, Rockefeller University, USA

CEO



Jaehyeon Ryu

- Global Expert in Commercialization
(Remicade, Stelara, Tremfya, Darzalex, Imbruvica, etc.)

Education and Career

- Specialty Business Unit Head, Janssen North Asia
- Asia Pacific Strategic Marketing, Janssen Asia Pacific
- Marketing Director, Xian Janssen
- Global Strategic Marketing, New Jersey, US
- Graduate of Seoul National University College of Pharmacy

Leadership

New Drug & Bio Research Center Senior Director / Wonil Lee	Central Research Center Director / Jungin Pyo	Clinical Development Director / Pankyung Kim	Business Development Senior Director / Jeonghan Kim	Strategic Planning Senior Director Jaejin Choi
- Ph.D. from Seoul National University College of Pharmacy - Director of New Drug Research Division at Daewon Pharmaceutical - Senior Researcher, New Drug Research, Jeil Pharmaceutical Co. - Director of Research and Development Division at GemVax & KAEI - Principal Researcher at C&C New Drug Research Institute	- Ph.D. in Medicinal Chemistry from Yonsei University - Team Leader at Intron Biotechnology, Inc. - Adjunct Professor at Sahmyook University - Postdoctoral Researcher at the Korea Institute of Science and Technology (KIST) (diabetes treatment, antihistamine treatment, urinary incontinence treatment)	- Master's degree in Medicine from Kyungpook National University - Clinical Team Leader at Yooyoung Pharmaceuticals - Clinical Team Leader at Hyundai Pharmaceutical - Clinical Team Leader at Hanmi Pharmaceuticals - Clinical Study Leader at LG Life Sciences	- MBA from UC Berkeley - Business Development and Global Marketing at Dongwha Pharmaceuticals - Team Leader at Celltrion in the Business Development Team - Marketing Team Leader at Bristol Myers Squibb Korea - Investment Analyst at Bayer, USA	- Master's degree in Business Administration from Sogang University - Head of Commercial Excellence at HanDok - Sales Director and Head of Commercial Excellence at Korea Elanco - Director of Strategic Planning and Marketing Excellence at Bayer Korea - Marketing Team Leader at Janssen Korea

1. Company Introduction

Pipeline Overview 1) Core Pipelines

Pipeline	Indication	Ad Route	Key Features	Development Status	Discovery	Pre-Clinical	Phase 1	Phase 2	Phase 3
CU06	Diabetic Macular Edema (DME)	Oral	- First in class oral therapeutics for retinal diseases - Demonstrated potential for vision improvement in DME patients.	Completed Phase 2a clinical trial in the U.S. (Apr 2024)					
CU01	Diabetic Nephropathy	Oral	- First-in-Class - demonstrated the potential to improve not only the uACR but also eGFR	Ongoing Phase 2b clinical trial in Korea (Patient enrollment completed)					
CU71	Neurovascular Disease	Oral	- First-in-Class - Next-Generation Endothelial Dysfunction Blocker	- Korea patent application filed (July 2024) - Preclinical studies in progress					
MT-101	Acute Kidney Injury, Chronic Kidney Disease	IV	- First-in-Class - Tie2-activating antibody	Preclinical studies in progress					
MT-103	DME, wAMD, Diabetic Retinopathy (DR)	IV	- First-in-Class - Tie2-activating X anti-VEGF	Preclinical studies in progress					

1. Company Introduction

Pipeline Overview 2) Emerging Pipelines

Pipeline	Indication	Ad Route	Key Features	Development Status	Discovery	Pre-Clinical	Phase 1	Phase 2	Phase 3
CU104	Ulcerative Colitis	Oral	• First-in-Class	• Completed Phase 1 clinical trial in the U.S					
CU106	Immuno-Oncology Combination	Oral	• First-in-Class	• Completed Phase 1 clinical trial in the U.S					
CU03	wAMD	Oral	• First-in-Class	• Completed Phase 2a clinical trial in Korea					
CP01-R01	Kidney Failure in Companion Animals	Oral	• First-in-Class • Confirmed therapeutic effects on kidney disease in animal models	• Clinical trial IND submitted (Aug 2024)					
MT-201	Anti-Thrombosis	IV	• First-in-Class • Novel Anti-coagulant	• Lead compound identified					
MT-202	Ischemic/ Reperfusion MI/Stroke/PT	IV	• First-in-Class • Novel Anti-coagulant X Tie2	• Lead compound identified • (Bi-specific antibody)					
MT-102	Critical Limb Ischemia (CLI)	IV	• First-in-Class	• Preclinical studies in progress					
MT-104	Solid Tumor	IV	• First-in-Class	• Discovery (Bi-specific antibody)					
MT-105	Inflammatory Bowel Disease (IBD)	IV	• First-in-Class	• Discovery (Tri-specific antibody)					
MT-40X	Solid Tumor	IV	• First-in-Class	• Discovery					

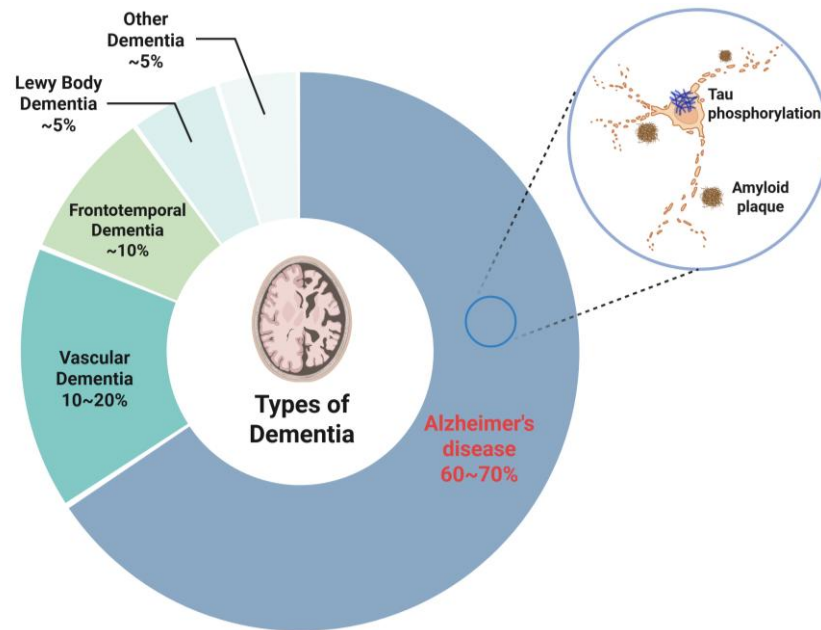
CU71

Targeting BBB Dysfunction:

A Novel Approach to Alzheimer's Therapy

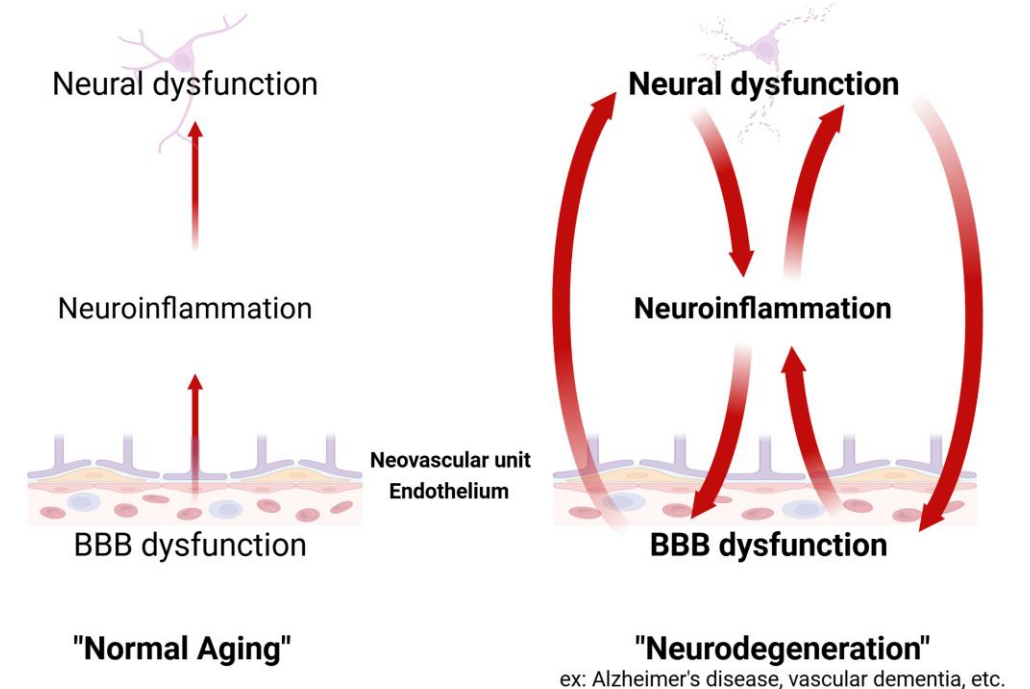
Alzheimer's Disease: The Central Role of BBB and Neuroinflammation

Types of dementia¹⁾



- **AD accounts for 60-70% of dementia cases** and often appears in mixed forms
- Key AD mechanisms include **amyloid-beta accumulation** and **tau phosphorylation**
- **BBB dysfunction** and **neuroinflammation** are emerging as shared causes across dementia types

BBB dysfunction and Neuroinflammation in aging and AD²⁾

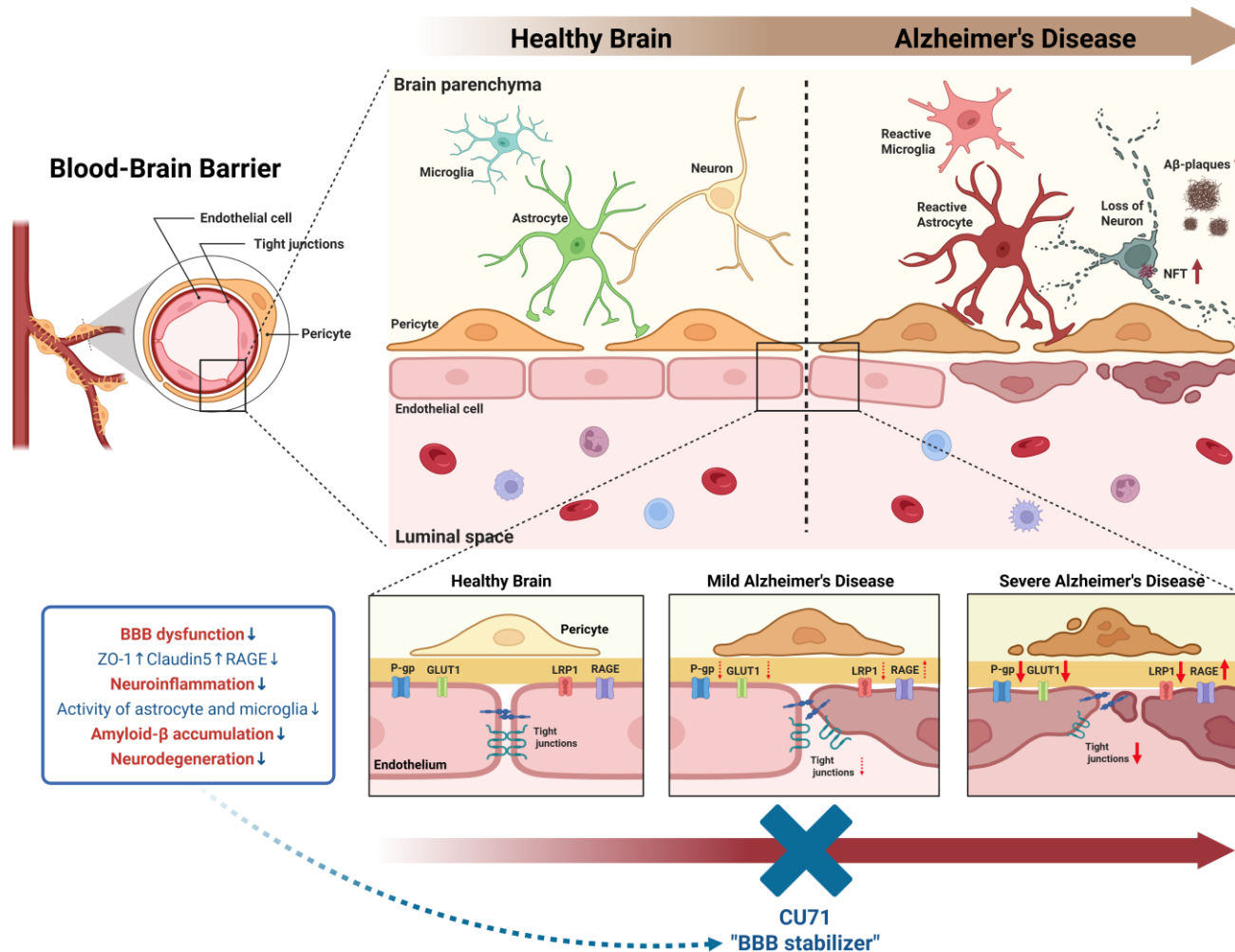


- **Normal aging:** BBB dysfunction → Neural dysfunction
- **Neurodegeneration:** BBB dysfunction, neuroinflammation, and neural dysfunction are interrelated and influence each other

1) Adapted from *Types of Dementia*. Alzheimer Calgary, <https://www.alzheimercalgary.ca/learn-about-dementia/types-of-dementia>. Accessed 28 May 2025. 2) Adapted from Stroke. 2023 Mar;54(3):661-672 doi: 10.1161/STROKEAHA.122.040578.

Pathophysiology of AD: BBB dysfunction & Neural dysfunction

BBB dysfunction and neuroinflammation is recognized as a common pathological phenomena of Alzheimer disease

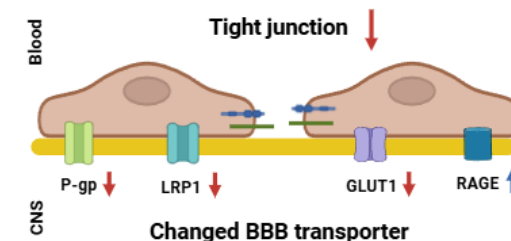


BBB dysfunction

- Disrupted tight junctions
- Increased BBB permeability
- Dysfunction of BBB transporters
- Inflammatory activation of ECs
- Reduced cerebral perfusion

Neural dysfunction

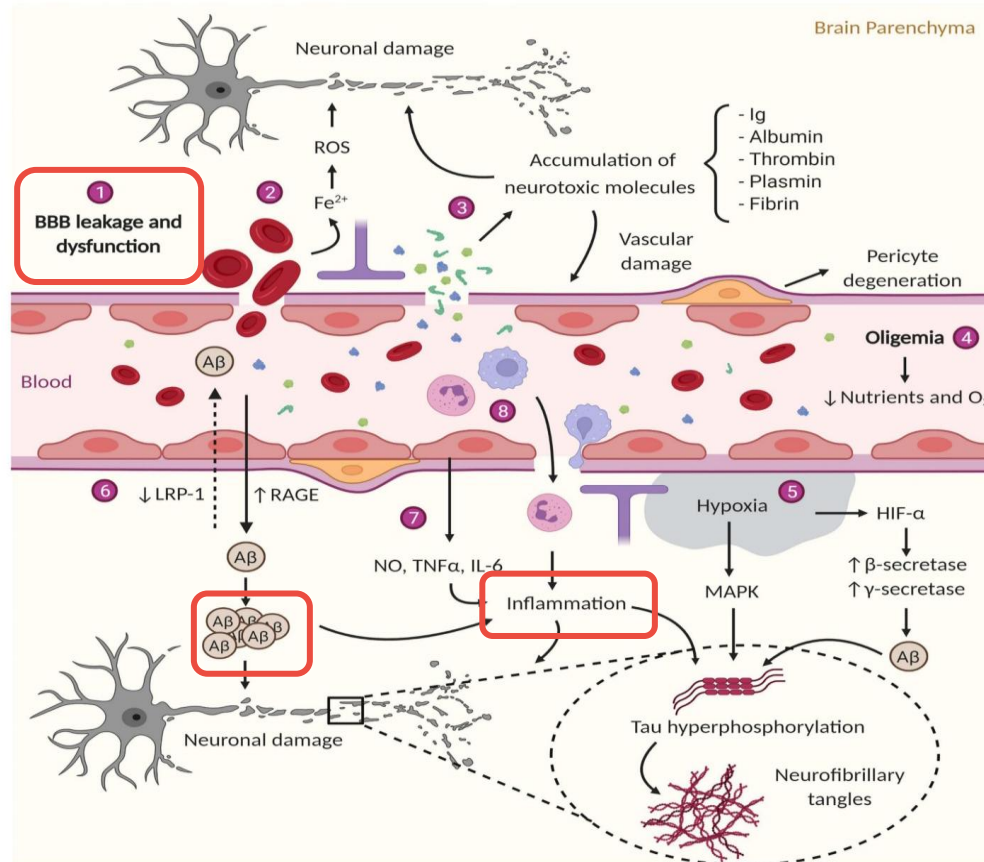
- Amyloid plaque and neurofibrillary tangle ↑
- Neuronal loss
- Synaptic dysfunction
- Memory loss and cognitive impairment



- **P-gp**: Reduced in AD → less Aβ clearance, more brain accumulation
- **LRP1**: Key for Aβ clearance → dysfunction increases brain Aβ
- **GLUT1**: Glucose transporter → reduced in AD/aging, lowers brain energy supply
- **RAGE**: Brings Aβ/AGEs into brain → triggers inflammation, worsens BBB damage

BBB Dysfunction & Neuroinflammation: Both a Cause and Aggravating Factor in AD

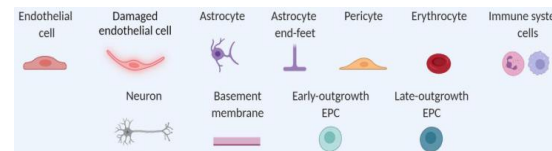
Alzheimer's Disease Capillary¹⁾



- (1) The **rupture and dysfunction of the BBB** increase the permeability of different cells and molecules
- (2) **Damage brain cells** through the production of ROS
- (3) **induce neurodegeneration and vascular damage**
- (4) **Reduce nutrient and oxygen supply**
- (5) **Increase Aβ production and tau phosphorylation**
- (6) **Accumulate Aβ in the brain**
- (7) **Release a large amount of proinflammatory mediators (NO, TNFα, and IL-6)**
- (8) **Extravasation of immune system cells generate inflammation**

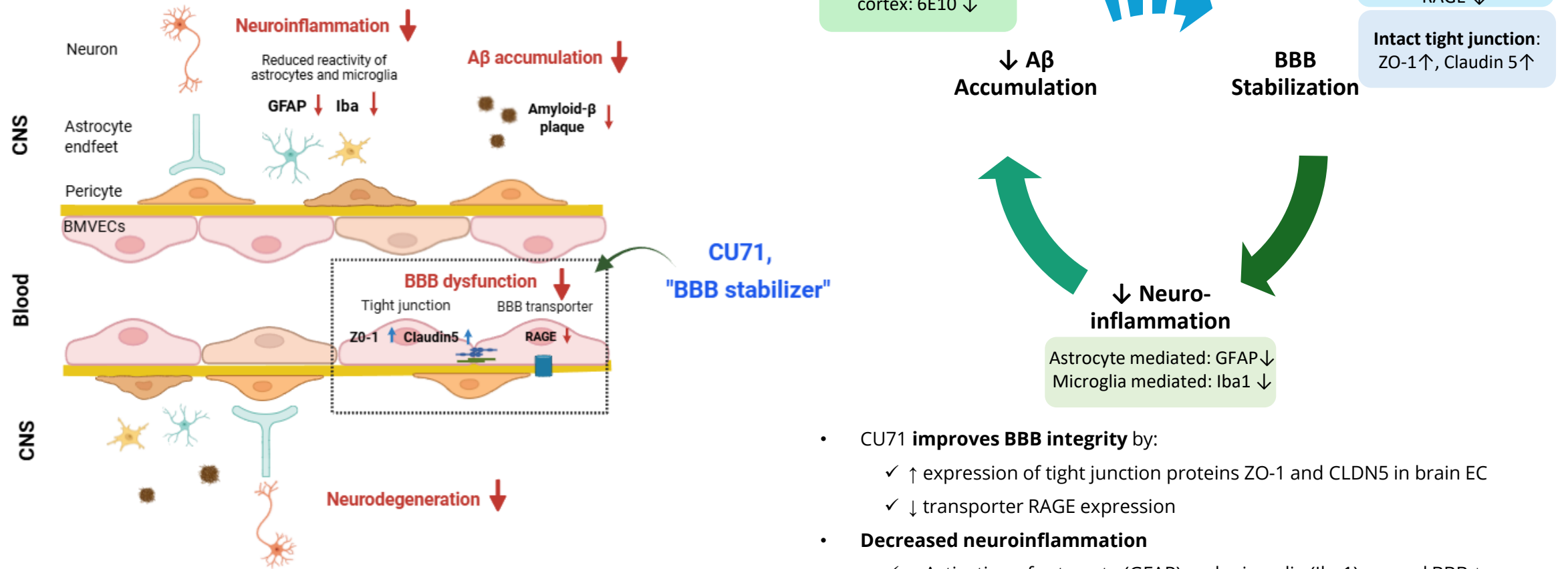


NEURONAL DAMAGE



BBB stabilizer treating neurodegenerative disease

CU71 ameliorates cognitive impairment by inhibiting BBB dysfunction and neuroinflammation in Alzheimer's disease



- CU71 **improves BBB integrity** by:
 - ✓ ↑ expression of tight junction proteins ZO-1 and CLDN5 in brain EC
 - ✓ ↓ transporter RAGE expression
- **Decreased neuroinflammation**
 - ✓ Activation of astrocyte (GFAP) and microglia (Iba1) around BBB ↓
- **Reduction in Aβ accumulation** → neurodegeneration suppressed

CU71

Results:

In Vivo Findings in Alzheimer's Disease

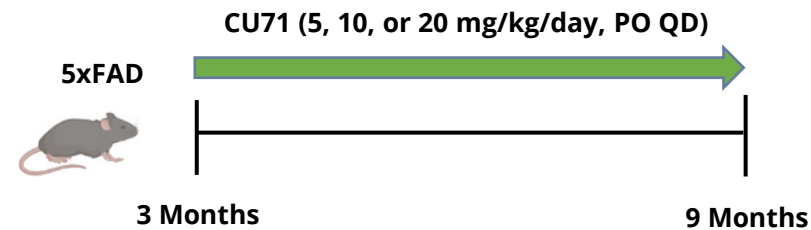
Preclinical Findings: CU71 Improves Cognitive Function in AD

Head-to-Head Comparison of CU71 and Donepezil in 5xFAD Mouse Model of Alzheimer's Disease

● Materials & methods

- Mouse: 5xFAD (3 months, male)
- CU71: 5, 10 or 20 mg/kg in Vehicle by PO QD
- Groups (n=12)

Group
G1: Wild type (WT) + Vehicle
G2: 5xFAD + Vehicle
G3: 5xFAD + Donepezil (1 mg/kg)
G4: 5xFAD + CU71-5 (5 mg/kg)
G5: 5xFAD + CU71-10 (10 mg/kg)
G6: 5xFAD + CU71-20 (20 mg/kg)



- Novel object recognition
- Morris water maze



Histological analysis (n=6)

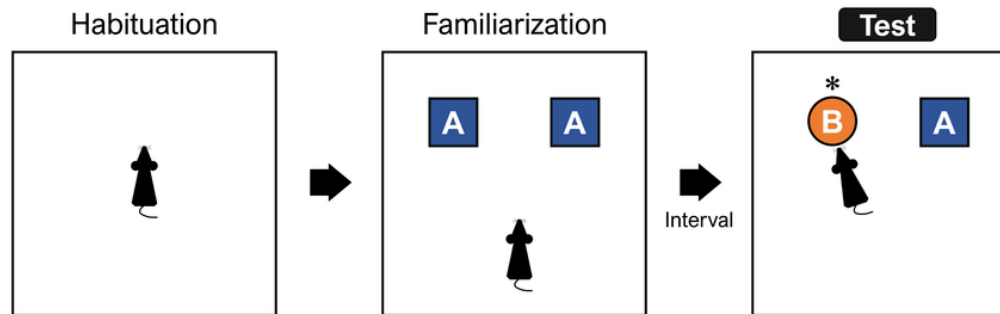
- Immunohistochemistry staining
- Western analysis

The results will be presented as a poster at Alzheimer's Association International Conference (AAIC) 2025

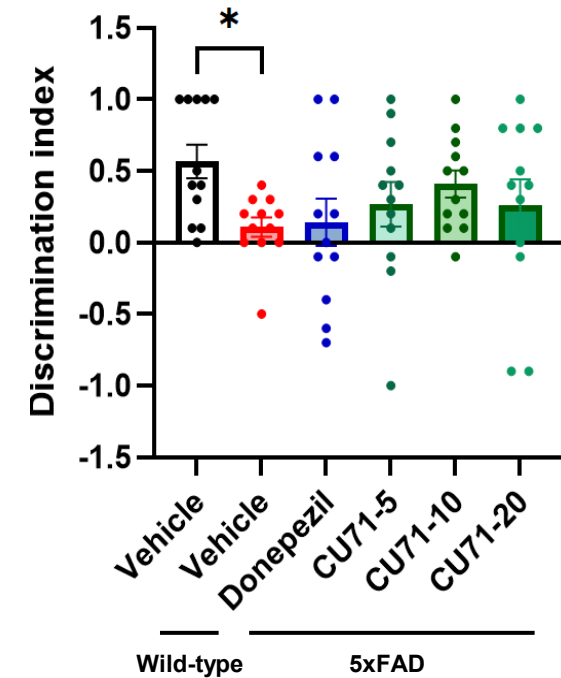
CU71 improves recognition memory in the 5xFAD mice

The novel object recognition (NOR) test evaluates the ability to recognize a new object

- Novel Object Recognition (NOR)



- In the experiment, 5xFAD mice showed significantly impaired cognition compared to wild-type mice.
- CU71 showed a dose-dependent trend toward cognitive improvement.



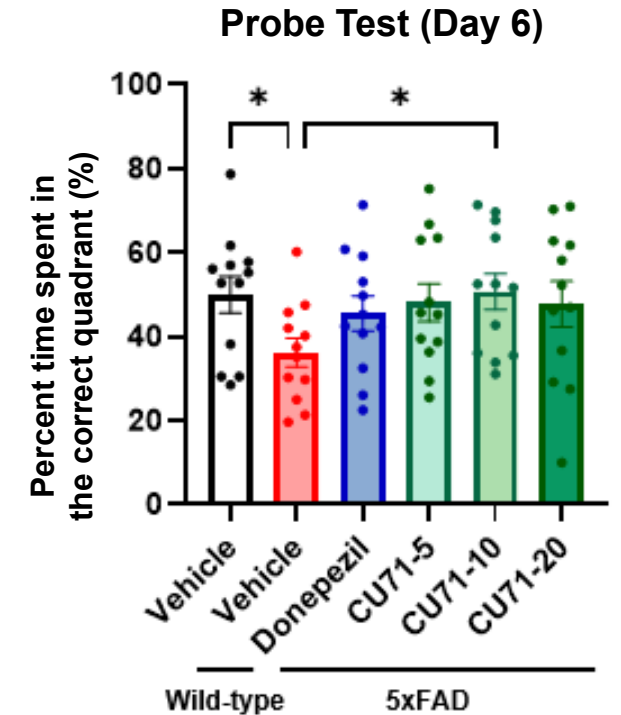
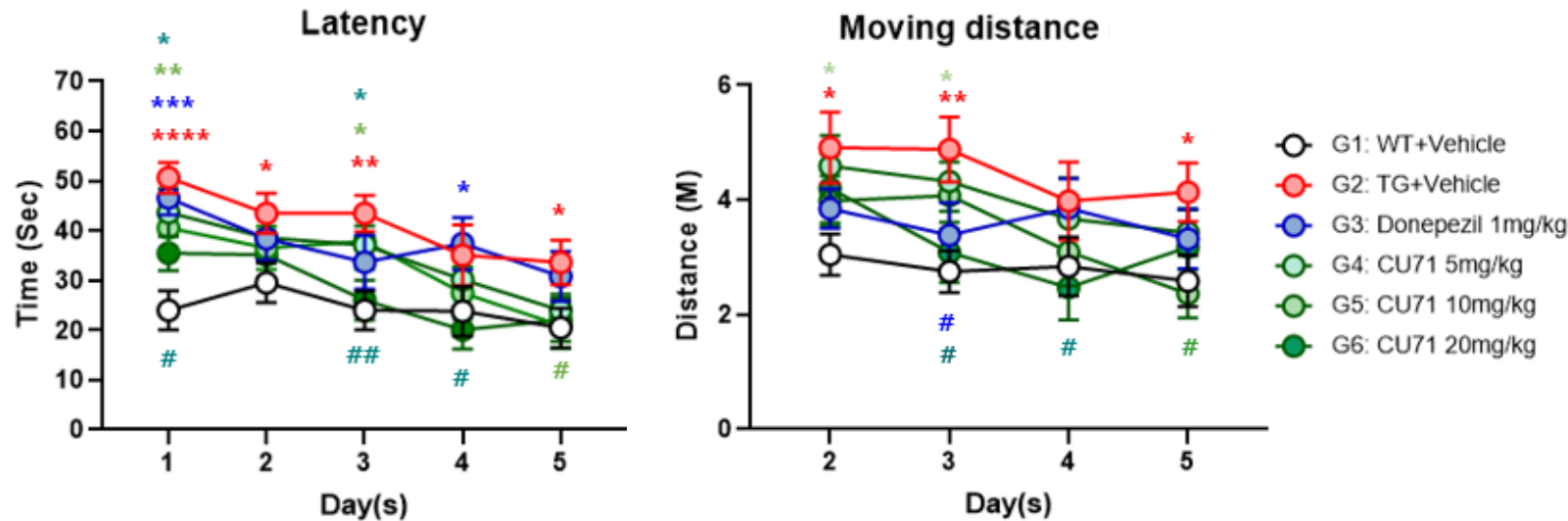
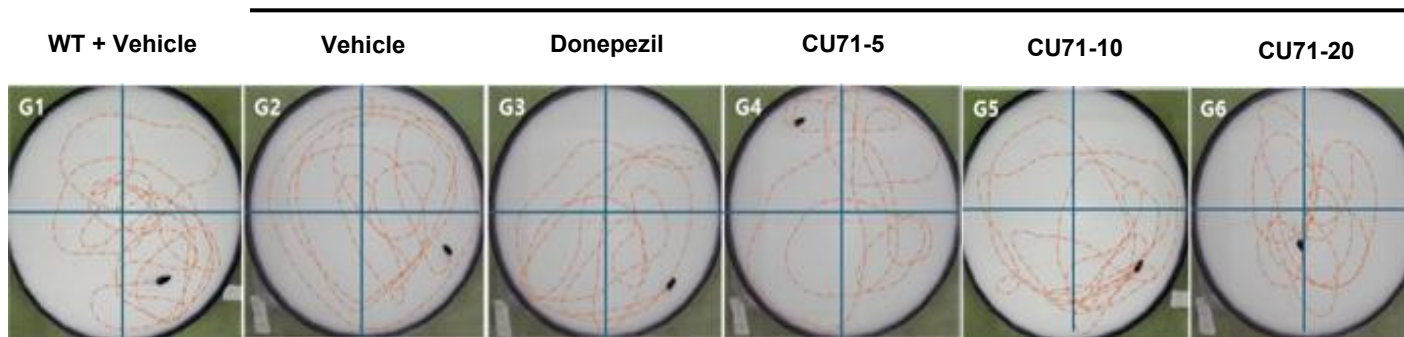
*: $p < 0.05$

CU71 improves spatial learning and recognition memory in the 5xFAD mice

The Morris water maze test assesses the ability to remember the location of a hidden platform in a water-filled pool

● Morris water maze (MWM)

5xFAD



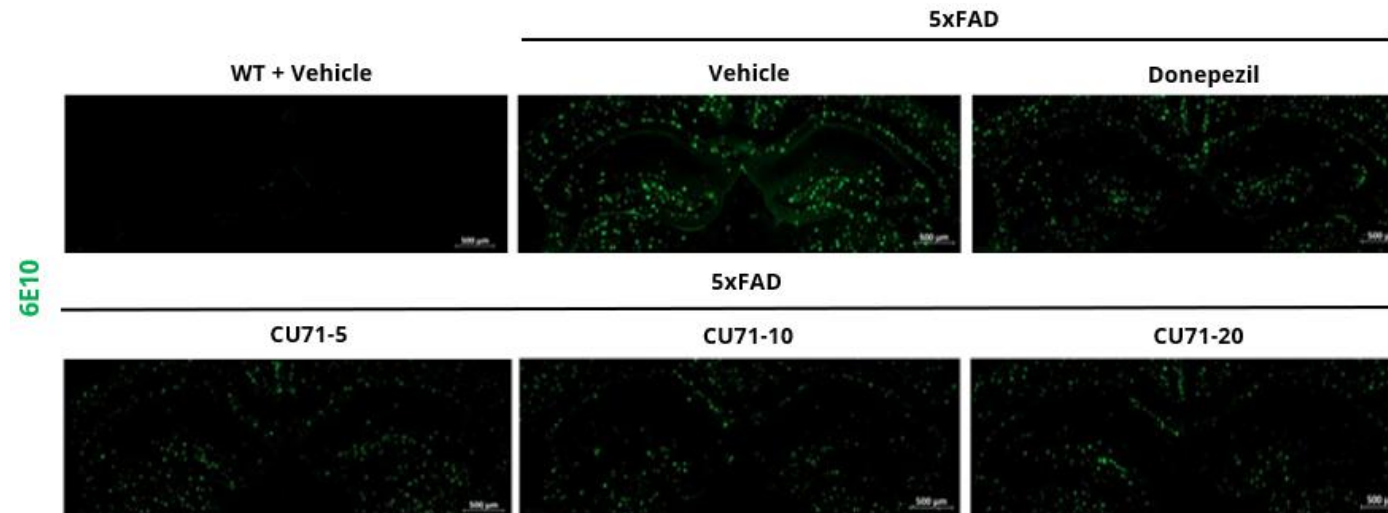
- * : $p < 0.05$, ** : $p < 0.01$, *** : $p < 0.001$; G1 vs All groups
- # : $p < 0.05$, ## : $p < 0.01$; G2 vs All groups

- Both latency and distance were significantly reduced on the fifth and final day in the CU71 10mg group.
- CU71 10mg group spent significantly more time in the correct quadrant in the probe test compared with the vehicle

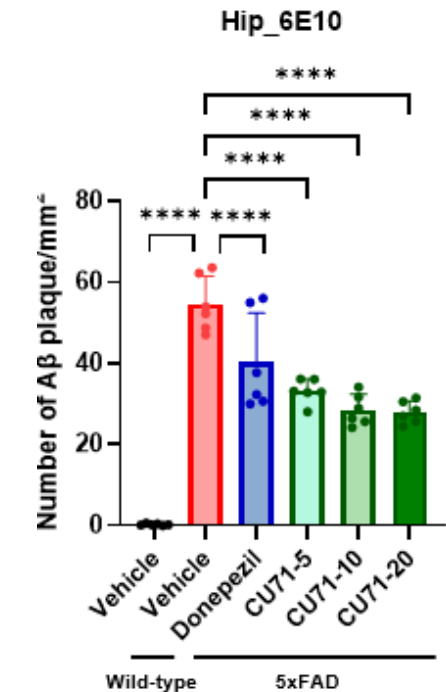
CU71 reduces A β deposition in the **hippocampus** of 5xFAD mice

The hippocampus is a critical region responsible for the formation and storage of memory

- Hippocampus



6E10: A marker of amyloid- β

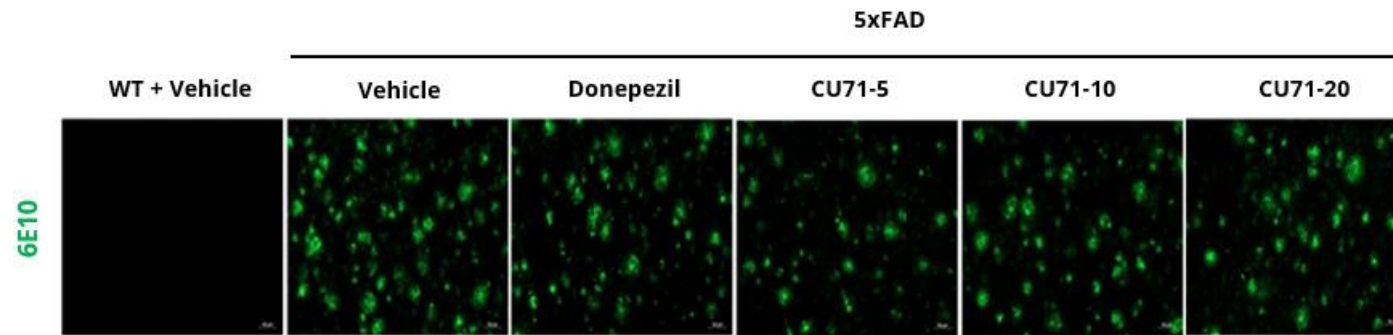


- CU71 significantly reduces A β plaque deposition in the hippocampus across all dose groups compared to the vehicle group

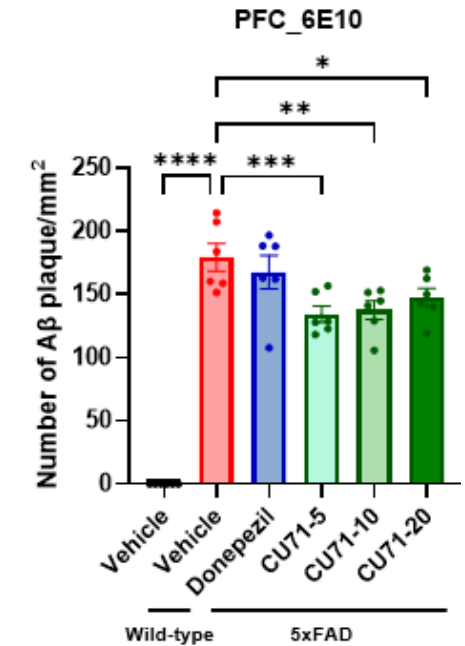
CU71 reduces A β deposition in the **cortex** of 5xFAD mice

Amyloid- β plaques spread from the hippocampus to the cortex in Alzheimer's disease

- Cortex**



6E10: A marker of amyloid- β

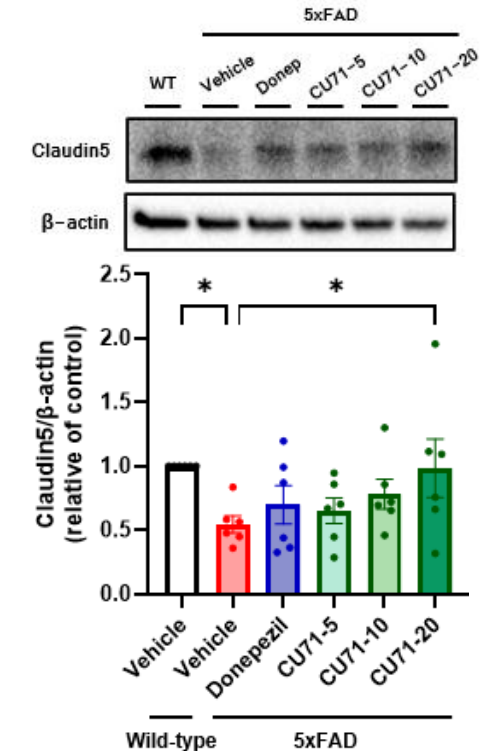
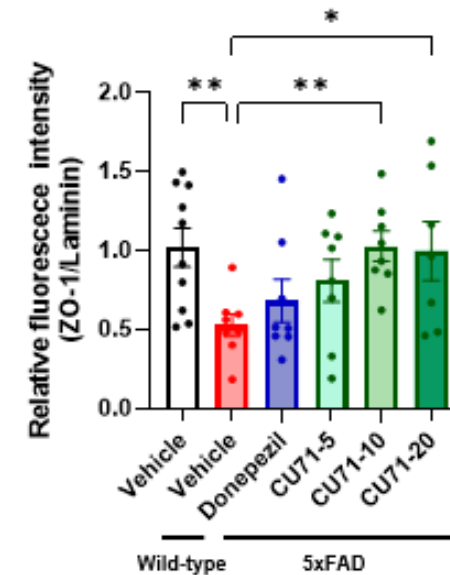
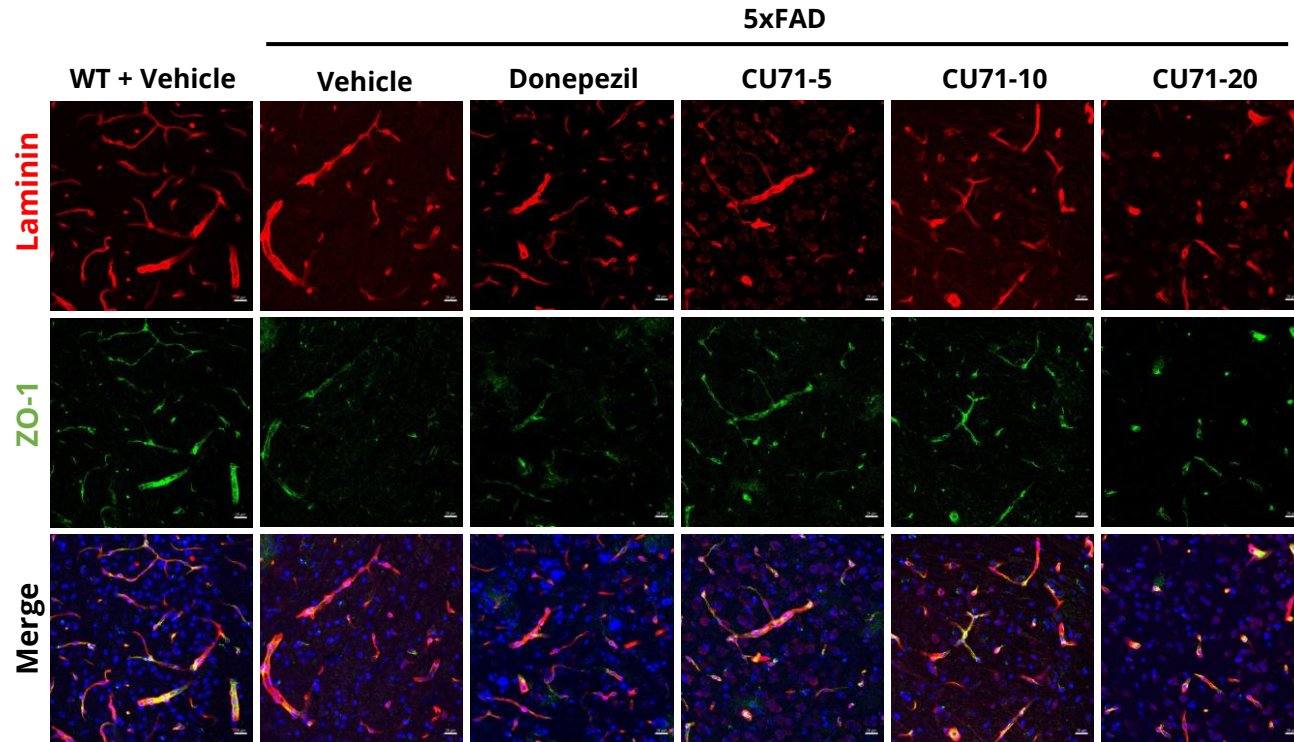


* : $p < 0.05$, ** : $p < 0.01$, *** : $p < 0.001$, **** : $p < 0.0001$

- CU71 significantly reduces A β plaque deposition in the cortex across all dose groups compared to the vehicle group

CU71 maintains BBB tight junctions in the 5xFAD mice

CU71 reduces BBB dysfunction by preserving endothelial cell junctions

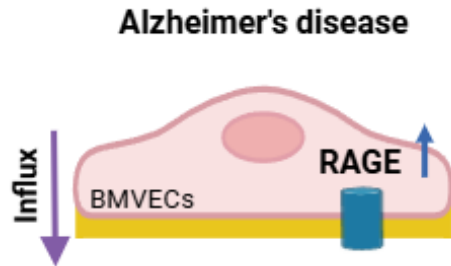


* : $p < 0.05$, ** : $p < 0.01$

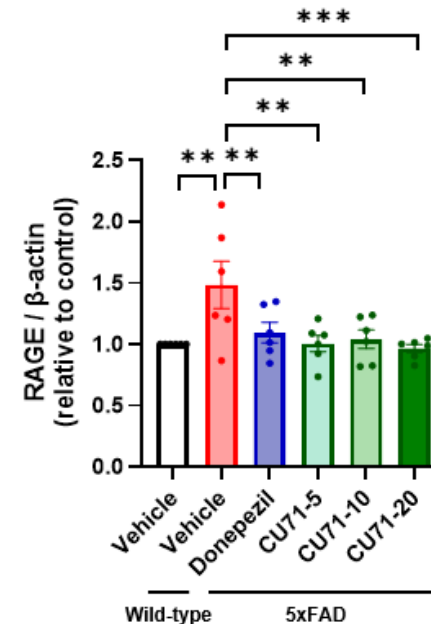
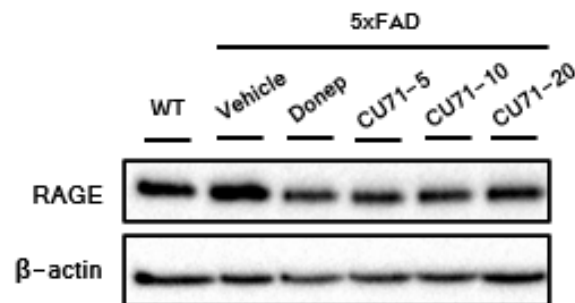
- **Laminin**: Vessel marker
- **ZO-1**: Zonular occludens-1, tight junction marker
- **Claudin5**: Tight junction protein, a marker of BBB integrity

- CU71 treatment results in significant dose-dependent increase in the expression of ZO-1 and Claudin-5, restoring their levels to those of normal mice

CU71 reduces RAGE expression in the 5xFAD mice



- RAGE (Receptor for Advanced Glycation End-products) is highly expressed in BMVECs.
- It plays a role in transporting amyloid- β from the blood into the brain.
- Its activation leads to BBB dysfunction, Neuroinflammation, and Neuronal damage.

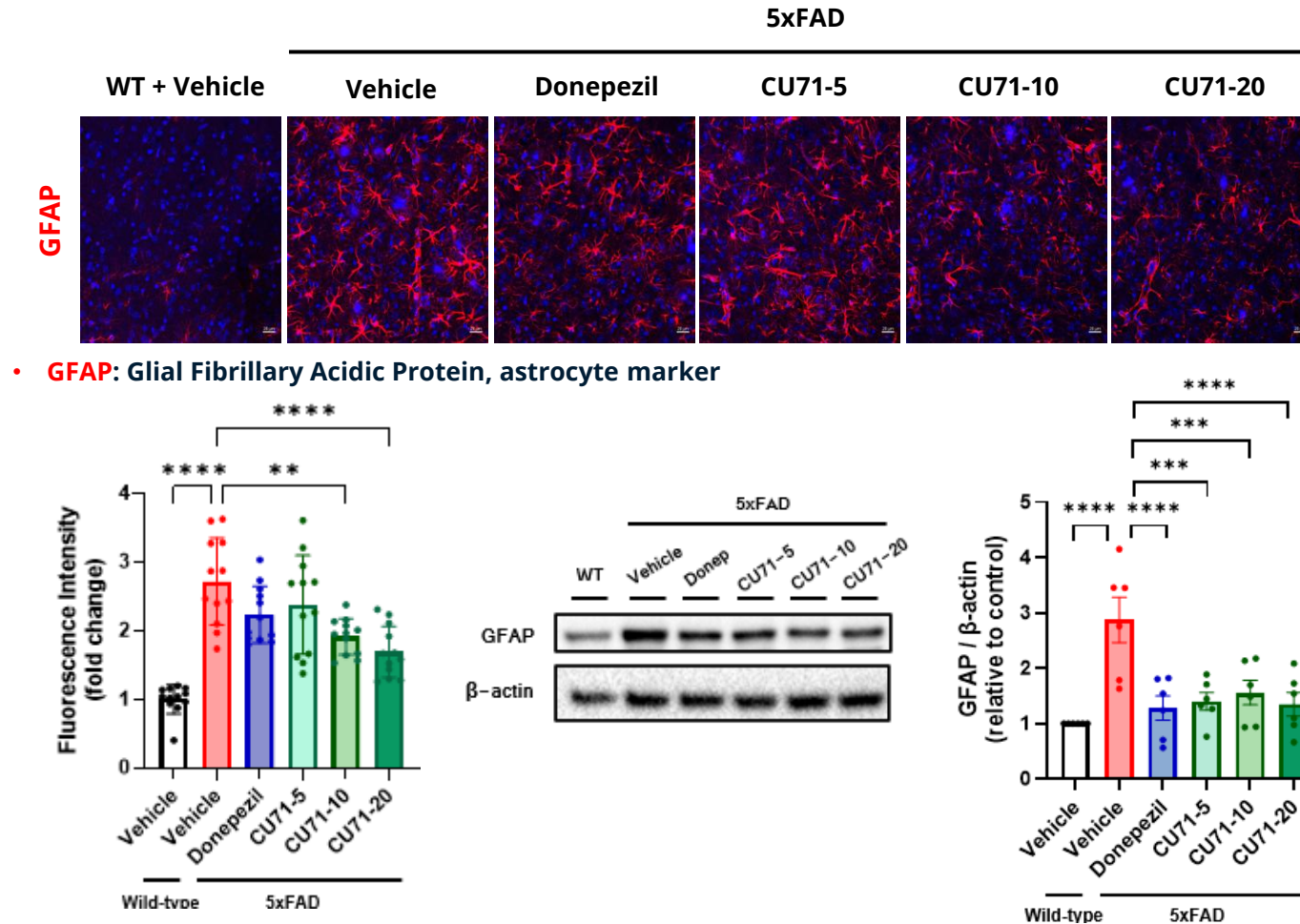


** : $p < 0.01$, *** : $p < 0.001$

- Protein expression analysis of RAGE in brain tissue showed a significant increase in the vehicle group, whereas all CU71-treated groups maintained RAGE expression at levels comparable to the normal group.

Reduced **astrocyte**-mediated neuroinflammation by CU71

Astrocyte reactivity to A β and microglia drives inflammation and neuronal damage



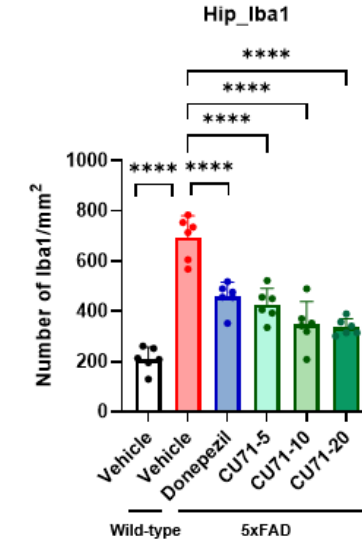
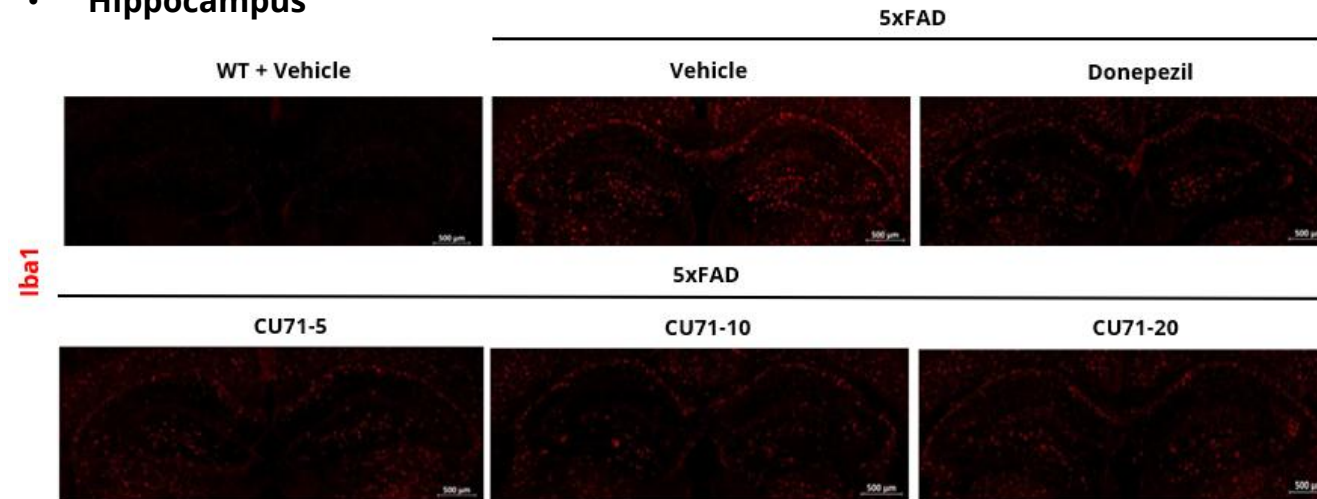
** : $p < 0.01$, *** : $p < 0.001$, **** : $p < 0.0001$

- GFAP is primarily expressed in astrocytes, with low levels in healthy brains.
- Overexpression of GFAP indicates astrocyte activation and neuroinflammation.
- Astrocyte activation is markedly increased in TG mice compared to WT.
- Significantly reduced GFAP levels observed in all treatment groups.

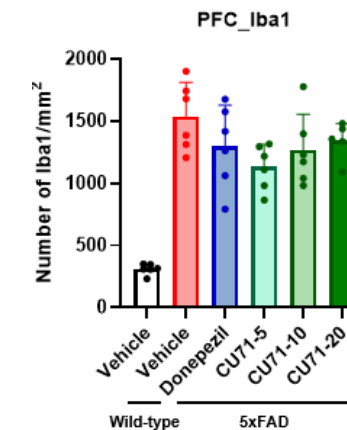
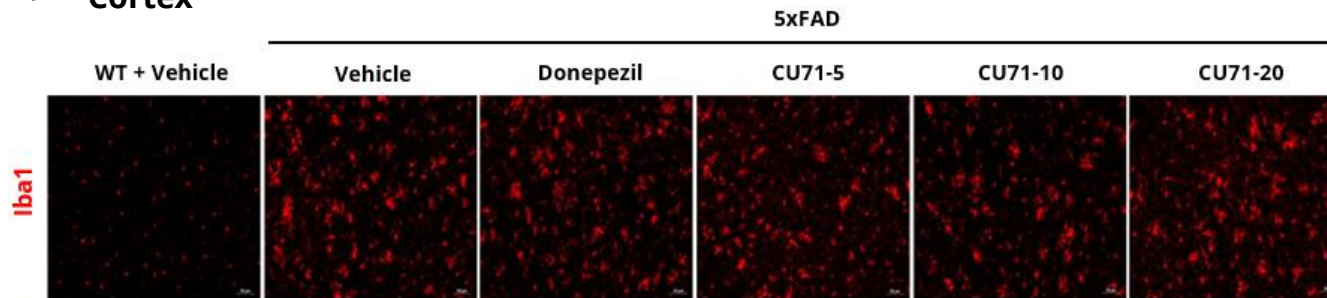
Reduced **microglia**-mediated neuroinflammation by CU71

Microglia trigger inflammation and neuronal damage in response to A β

• Hippocampus



• Cortex

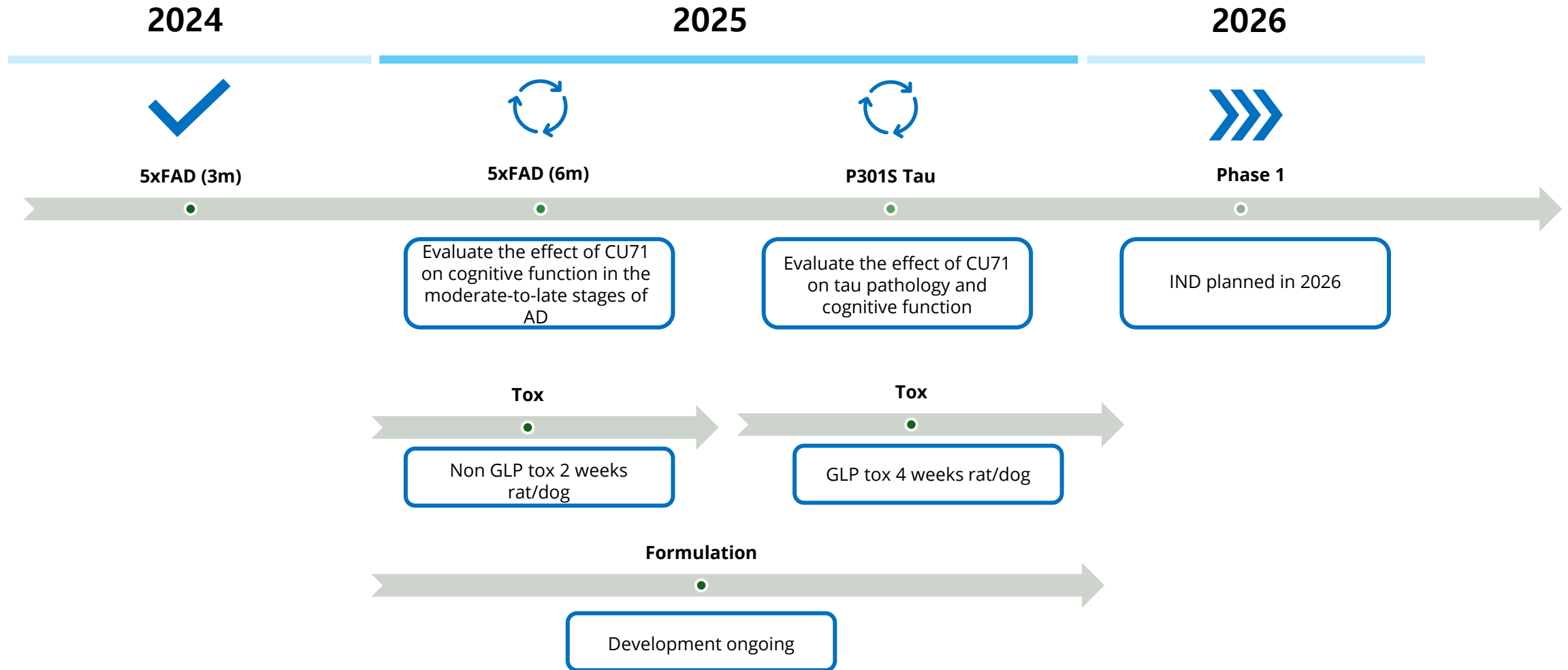


Iba1=Ionized calcium-binding adaptor molecule 1, a marker evaluating microglial activation

- CU71 significantly reduces Iba1 expression across all dose groups, indicating CU71's inhibition on microglial activation

* : $p < 0.05$, **** : $p < 0.0001$

Development Status & Next Steps



Key takeaways

[Key Features]

- A novel Alzheimer's therapy targeting BBB dysfunction
- CU71 reduces BBB dysfunction by restoring BBB tight junction and normalizing BBB transporter, thereby reducing neuroinflammation and amyloid- β accumulation

[Key findings in 5xFAD model]

CU71 demonstrated greater cognitive improvement than donepezil in 3-month-old 5xFAD mice

- NOR test: Dose-dependent enhancement in recognition memory
- MWM test: Significant improvement in spatial learning and memory
- Amyloid- β deposition decreased in both the hippocampus and cortex
- ZO-1 and Claudin-5 expression restored to wild-type levels
- RAGE expression remained comparable to wild-type levels
- Astrocyte activation reduced, as indicated by decreased GFAP
- Microglial activation also suppressed by CU71 treatment

[Next Preclinical Development]

- **5xFAD (6 months old):** To evaluate CU71's effect in moderate-to-late stages of AD
- **P301S Tau transgenic mouse model:** To assess CU71's efficacy on tau pathology and cognition

[Clinical Development Plan]

- Phase I IND submission anticipated in 2026

[Patent Status]

- Compound patent: Filed in 2024
- Usage patent: Filed in 2025

[Partnership]

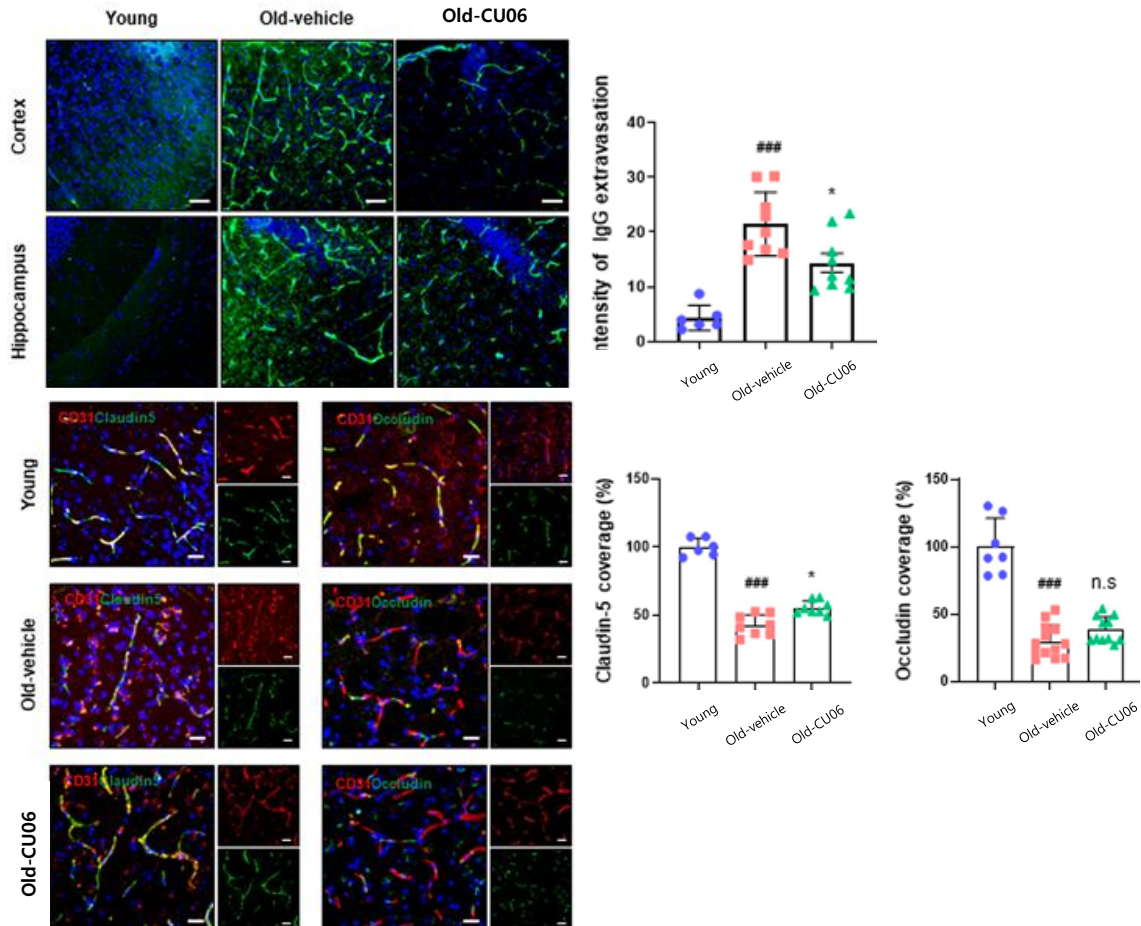
- Actively pursuing out-licensing worldwide

Development background of CU71: Brain

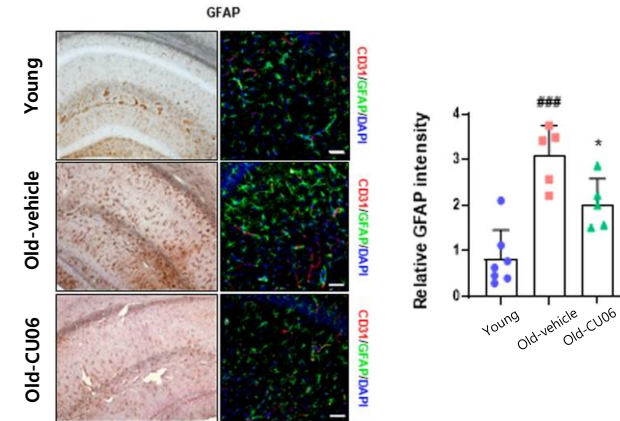
Neuroprotective effects in the Brain confirmed by another endothelial stabilizer (CU06)

❖ Aging model

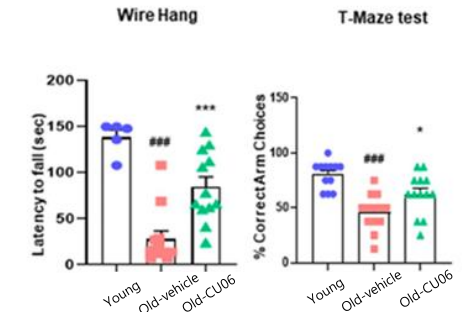
- Endothelial dysfunction rescued impaired BBB integrity and reduced tight junction protein coverage in aged mice



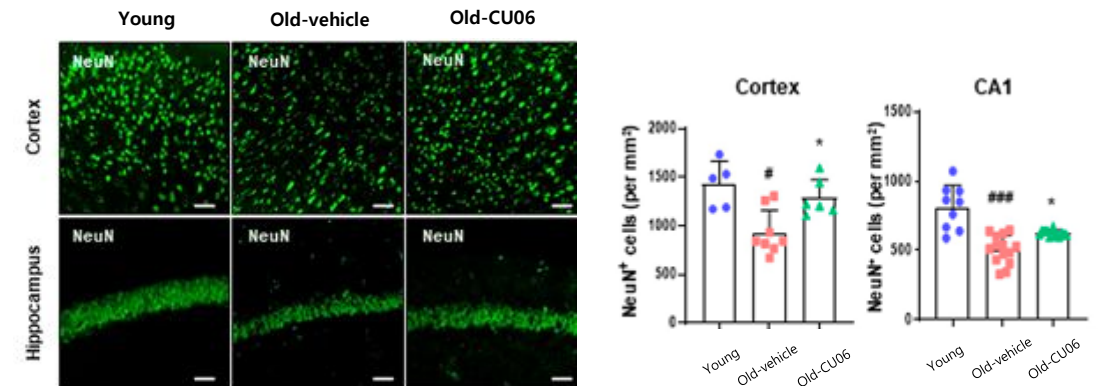
➤ Neuroinflammatory effects



➤ Cognitive function improvement effects



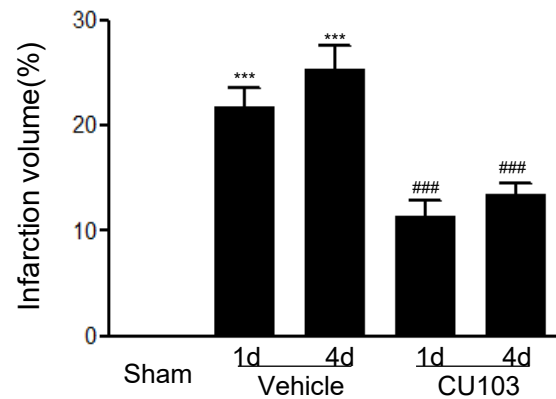
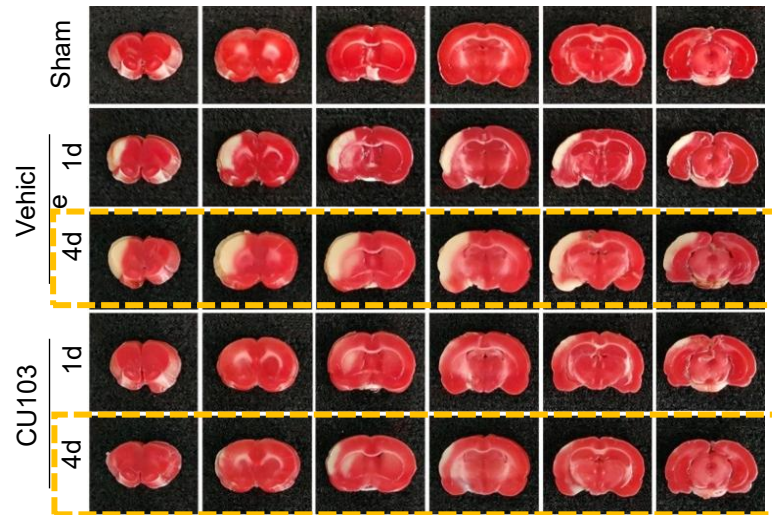
➤ Reduced the neuronal loss



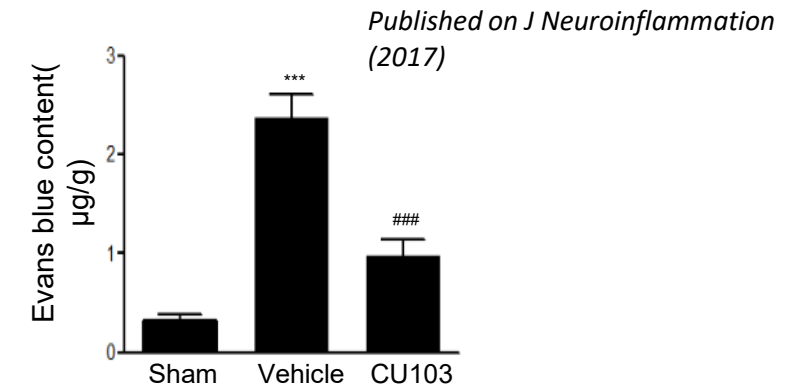
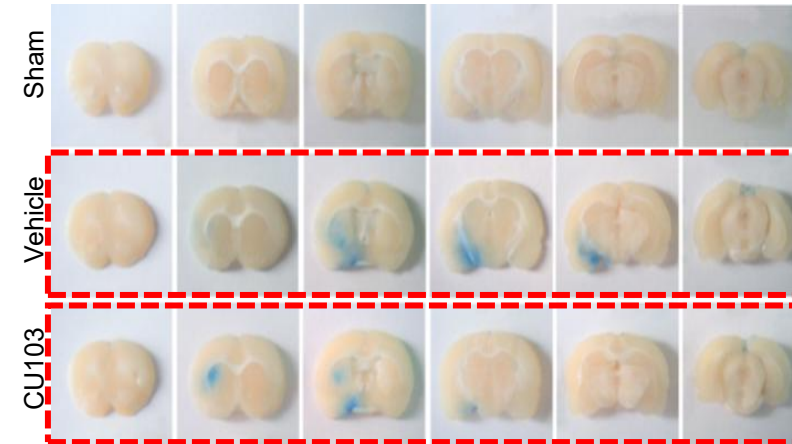
Development background of CU71: Stroke

Reduction in damage caused by I/R by inhibiting blood vessel leakage in brain confirmed by another endothelial stabilizer (CU06)

**Reduced infarct size
in brain damage caused by I/R**



Inhibition of cerebrovascular leakage



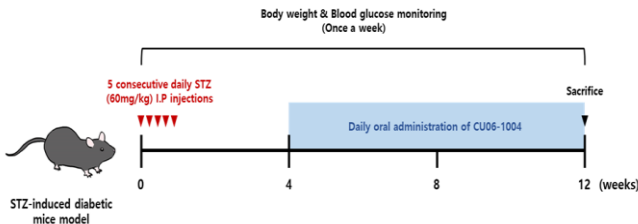
Published on J Neuroinflammation (2017)

Development background of CU71: Retina

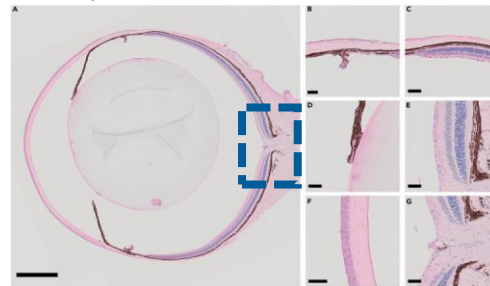
Neuroprotective effects via vascular normalization confirmed by another endothelial stabilizer (CU06)

Experimental Scheme

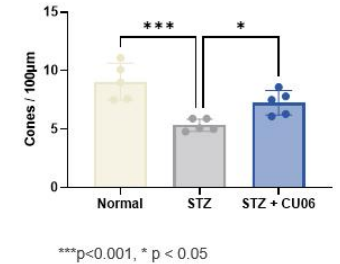
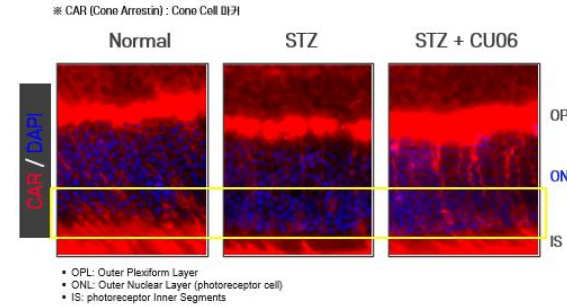
1) STZ-induced diabetic mouse model



2) Histological analysis of structural changes in the centra retina

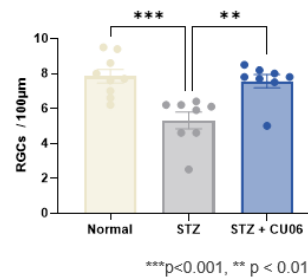
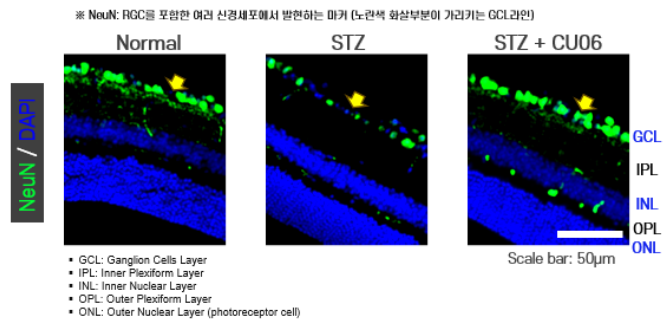


CU06 mitigates cone photoreceptor degeneration



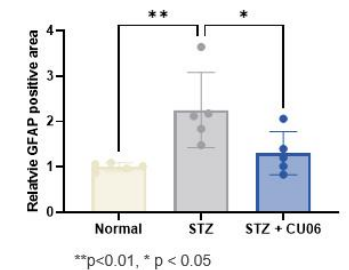
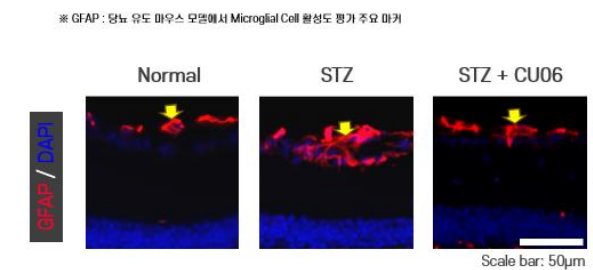
➤ Cone cell count **increased by ~36%** in CU06-treated group vs. STZ group

CU06 prevents retinal ganglion cell loss



➤ Ganglion cell count **increased by ~42%** in CU06-treated group vs. STZ group

CU06 modulates pathological reactive gliosis

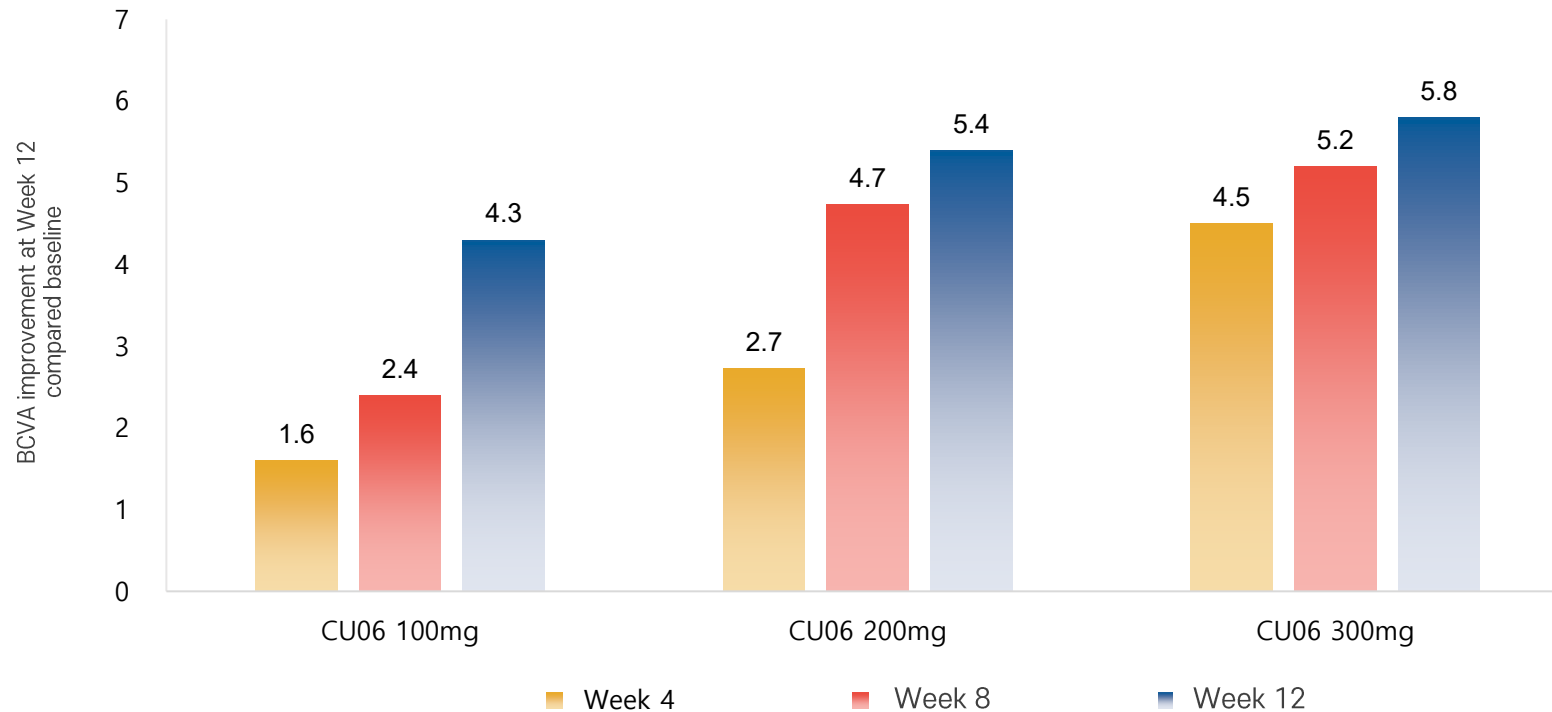


➤ Microglial cell count **decreased by ~42%** in CU06-treated group vs. STZ group

Development background of CU71: Retina (Phase 2a study in human)

Dose-dependent & treatment duration dependent BCVA improvement in patients with diabetic macular edema (DME)

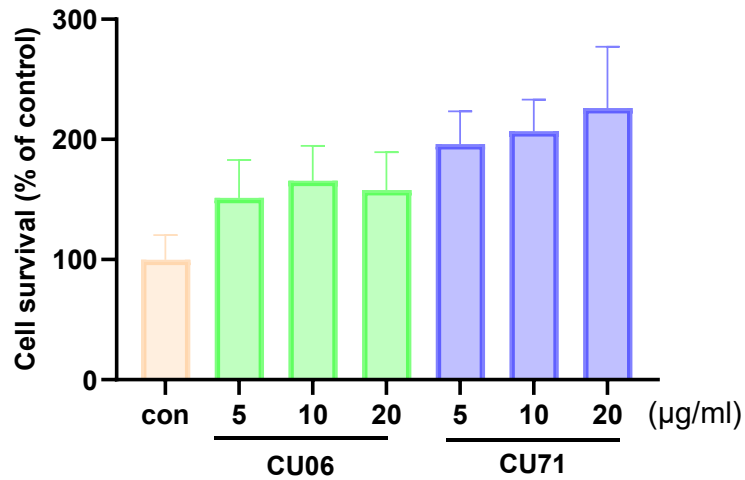
Subgroup Analysis: Improvement by dose and treatment duration in the patient group with BCVA≤69*



Data on file. CURACLE *Subgroup BCVA≤69 was prespecified on the Statistical Analysis Plan (SAP)

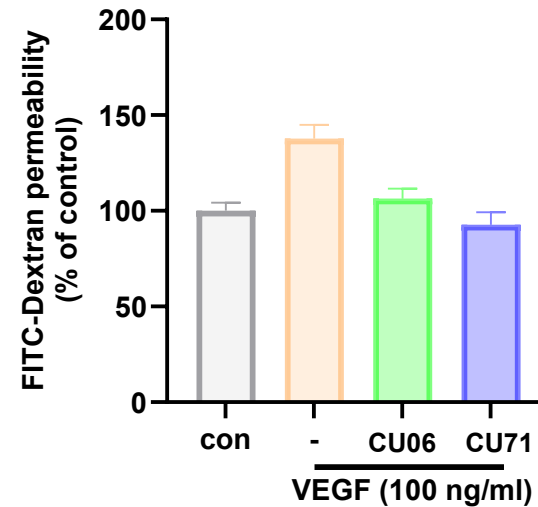
In Vitro Comparison of CU71-2021 and CU06-1004

Cell survival- ATP assay



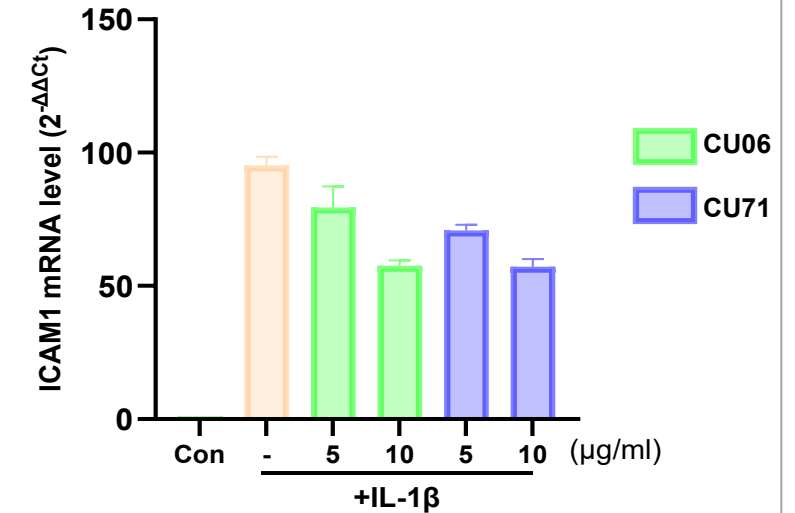
- Both 1004 and 2021 increased cell viability under serum-starvation conditions in a dose-dependent manner
- At equivalent concentrations, 2021 tended to show greater efficacy compared to 1004

Permeability assay

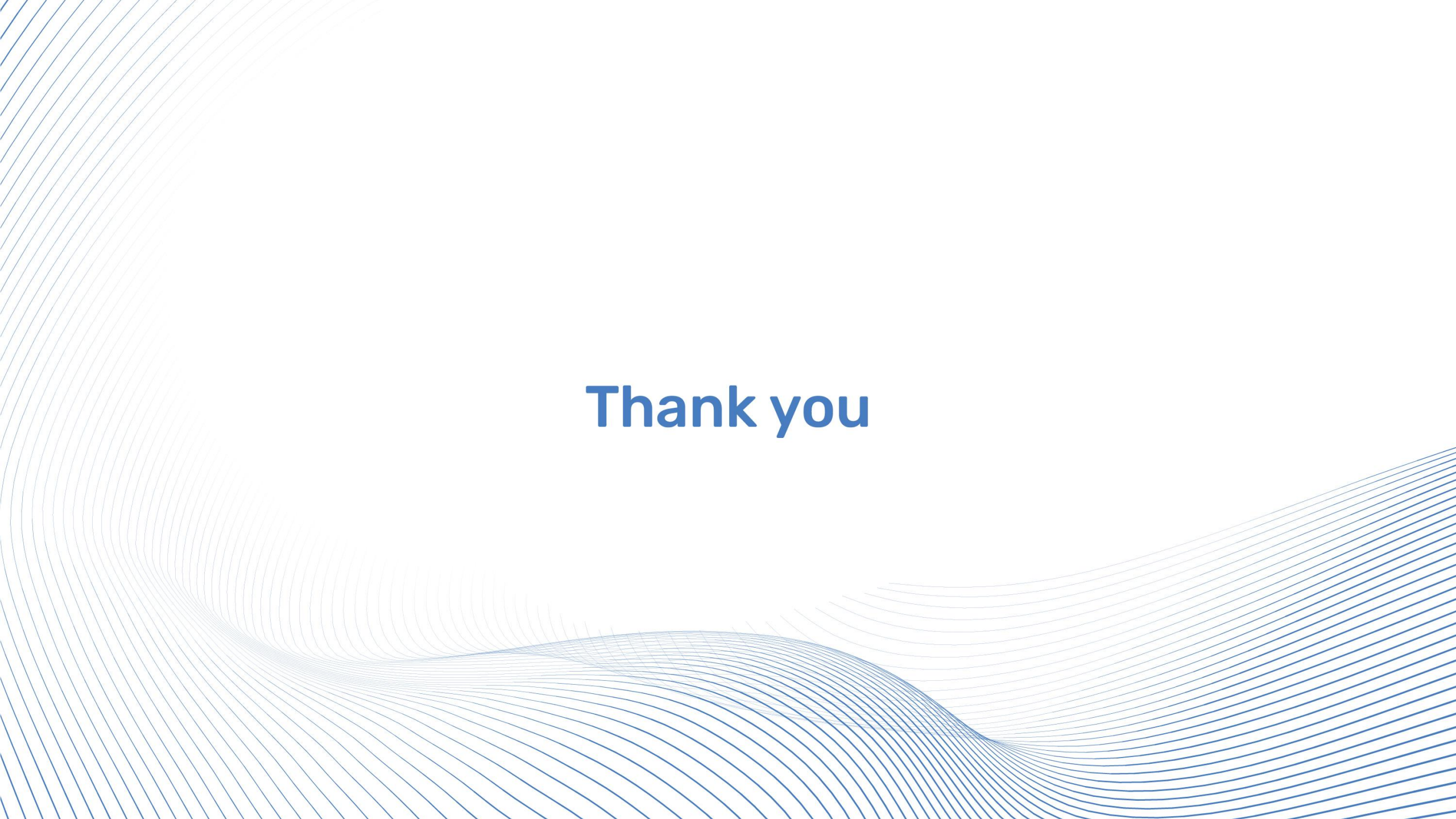


- Both 1004 and 2021 demonstrated inhibitory effects on VEGF-induced vascular leakage.

Anti-inflammation assay



- Both 1004 and 2021 reduced IL-1β-induced ICAM-1 expression in a dose-dependent manner.

The background features a series of thin, blue, wavy lines that create a sense of movement and depth. These lines are concentrated in the lower half of the image, forming a series of peaks and valleys that resemble a stylized landscape or a series of ripples in water. The lines are more densely packed in some areas, creating a darker blue, and more spread out in others, creating a lighter blue. The overall effect is a modern, minimalist, and dynamic background.

Thank you